Mathematical Derivation of Bistable Mechanisms for Vibration-Triggered Hook Deployment

Rolly Robot — Hook Bistable Actuator Simulations

Technical note for the Rolly hook-deployment simulations

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Contents

1 Introduction

A *bistable* mechanical system has two stable equilibrium positions separated by an energy barrier. If it is not disturbed, it stays in whichever well it already occupies. Moving from one well to the other requires enough input energy to cross the intervening saddle; the rapid transition is the *snap-through* event [4, 6].

For the Rolly hook, the first well holds the hook retracted. A RobStride RS05 quasi-direct-drive motor drives the surrounding structure with a sinusoidal force or displacement. Near resonance, the hook motion grows until the relative coordinate crosses the saddle. After snap-through the second well holds the hook deployed without continuous motor power. The model here is therefore concerned with three quantities: the barrier height, the natural frequency about the retracted state, and the actuator amplitude needed to reach the saddle.

The physical model derived and simulated here is **Simulation 2 (Physical)**: a compliant curved-beam bistable mechanism following the geometry of Qiu et al. [4] and the 3D-printed Design 9 of Zolfagharian et al. [5], with a companion two-degree-of-freedom tuned mass damper (TMD) model [2].

2 Compliant Curved-Beam Bistable Mechanism

2.1 Beam Geometry — Qiu Cosine Profile

Qiu et al. [4] proposed the initial (stress-free) shape of the curved beam as the first buckling mode of a fixed-end Euler beam, a cosine profile:

$$y_0(\xi) = \frac{h}{2}\left[1 - \cos\left(\frac{2\pi\xi}{l}\right)\right], \quad 0 \le \xi \le l \tag{1}$$

where h is the apex height [m] and l is the horizontal span [m]. The beam thickness is t [m], and the material elastic modulus is E [Pa].

Key geometric parameters:

$$Q = \frac{h}{t} \quad \text{(apex-height-to-thickness ratio)} \tag{2}$$

$$I = \frac{t^3}{12} \quad \text{(second moment of area per unit width)} \tag{3}$$

2.2 Bistability Condition

For a curved beam to exhibit bistability, two design conditions matter most [4, 5]:

1. **Q-ratio condition**: The apex height must substantially exceed the beam thickness:

$$Q = \frac{h}{t} \gtrsim 2.35 \tag{4}$$

   Qiu et al. report that a second-mode-constrained curved beam becomes bistable once this geometric ratio is large enough; their analysis and force-displacement plots put the threshold at about $Q = 2.35$ [4]. The MATLAB model uses $Q > 2.31$ as a numerical guard, but the more conservative design check is $Q \gtrsim 2.35$.

2. **Mode-2 suppression**: For a single beam, buckling mode 2 is asymmetric and prevents robust bistability. In the Zolfagharian Design 9, a *double-curved beam* connected by a central membrane at mid-span suppresses mode 2, so the snap-through path follows the symmetric constrained-beam behavior described by Qiu et al. [4].

2.3 Force-Displacement Relationship

2.3.1 Strain Energy Formulation

Using the Euler–Bernoulli beam assumptions, let the initial shape be $y_0(\xi)$ and the transverse displacement from that shape be $w(\xi)$, so $y(\xi) = y_0(\xi) - w(\xi)$ where w is the transverse displacement relative to the initial shape. The bending strain energy is then [4]:

$$U_b = \frac{EI}{2} \int_0^l \left(\frac{\mathrm{d}^2 w}{\mathrm{d}\xi^2} \right)^2 \mathrm{d}\xi \tag{5}$$

The axial compressive strain energy (due to shortening of the beam's projected length) is:

$$U_a = \frac{EA}{2l} \left(\int_0^l \frac{\mathrm{d}y_0}{\mathrm{d}\xi} \frac{\mathrm{d}w}{\mathrm{d}\xi} \, \mathrm{d}\xi \right)^2 \tag{6}$$

where $A = t \cdot b$ is the cross-sectional area (for width b).

The force-displacement relationship follows from virtual work: $F = \partial(U_b + U_a)/\partial d$, where d is the mid-point displacement (snapping direction).

2.3.2 Normalized Force-Displacement

Qiu et al. [4] show that the force-displacement curve depends only on Q and the normalized displacement $\bar{d} = d/h$. For the double-beam (mode-2 suppressed) configuration, the curve has the characteristic S-shape:

- **State 1** ($\bar{d} = 0$): zero force (equilibrium).
- **Peak push force** F_{push} at $\bar{d} \approx 0.27$: maximum force required to initiate snap-through.
- **Saddle** ($\bar{d} = \bar{d}_s$): force returns to zero (unstable equilibrium).
- **Valley force** $F_{\text{pop}} < 0$ at $\bar{d} \approx 0.82$: mechanism pulls itself toward State 2.
- **State 2** ($\bar{d} = 1$): zero force (second equilibrium).

2.3.3 Cubic Polynomial Fit

The simulation uses a cubic fit for the applied force in the compression-test convention. This is not a replacement for Qiu's modal solution or Zolfagharian's FEA; it is a compact surrogate with the correct equilibrium points and peak force:

$$F_{\text{ext}}(d) = A\, d\, (d - d_s)\, (d - d_f) \tag{7}$$

The coefficient A is calibrated so the force at the chosen peak location matches the Design 9 force level from Zolfagharian et al. [5]:

$$A = \frac{F_{\text{push}}}{d_{\text{push}}\, (d_{\text{push}} - d_s)\, (d_{\text{push}} - d_f)} \tag{8}$$

where $d_{\text{push}} \approx 0.27\, d_f$ is the location of the force peak.

The *internal restoring force* on an attached mass (Newton's third law) is:

$$F_{\text{restore}}(d) = -F_{\text{ext}}(d) = -A\, d\, (d - d_s)\, (d - d_f) \tag{9}$$

2.4 Potential Energy (Curved-Beam Model)

Integrating the restoring force analytically:

$$\begin{aligned} V(d) &= -\int_0^d F_{\text{restore}}(\xi)\,\mathrm{d}\xi = A\int_0^d \xi\,(\xi-d_s)\,(\xi-d_f)\,\mathrm{d}\xi \\ &= A\int_0^d \left[\xi^3-(d_s+d_f)\xi^2+d_s d_f\,\xi\right]\mathrm{d}\xi \\ &= A\left[\frac{d^4}{4}-\frac{(d_s+d_f)}{3}d^3+\frac{d_s d_f}{2}d^2\right] \end{aligned} \tag{10}$$

The energy barrier from State 1 to the saddle is:

$$\Delta E_1 = V(d_s)-V(0) = A\left[\frac{d_s^4}{4}-\frac{(d_s+d_f)}{3}d_s^3+\frac{d_s d_f}{2}d_s^2\right] \tag{11}$$

2.5 Natural Frequencies at Each Stable State

Linearizing $F_{\text{restore}}(d)$ about each equilibrium:

State 1 ($d=0$):

$$k_{\text{eff},1} = -\left.\frac{\mathrm{d}F_{\text{restore}}}{\mathrm{d}d}\right|_{d=0} = A\,d_s\,d_f \tag{12}$$

$$\omega_{n,1} = \sqrt{\frac{k_{\text{eff},1}}{m}}, \qquad f_{n,1} = \frac{1}{2\pi}\sqrt{\frac{A\,d_s\,d_f}{m}} \tag{13}$$

State 2 ($d=d_f$):

$$k_{\text{eff},2} = -\left.\frac{\mathrm{d}F_{\text{restore}}}{\mathrm{d}d}\right|_{d=d_f} = A\,d_f\,(d_f-d_s) \tag{14}$$

$$\omega_{n,2} = \sqrt{\frac{k_{\text{eff},2}}{m}}, \qquad f_{n,2} = \frac{1}{2\pi}\sqrt{\frac{A\,d_f\,(d_f-d_s)}{m}} \tag{15}$$

Since $d_s < d_f$ by definition, and typically $d_s \approx 0.55\,d_f$: $k_{\text{eff},2} = A\,d_f(d_f-d_s) \approx 0.45\,A\,d_f^2$, while $k_{\text{eff},1} = A\,d_s\,d_f \approx 0.55\,A\,d_f^2$, so $k_{\text{eff},2} < k_{\text{eff},1}$ and thus $f_{n,2} < f_{n,1}$.

2.6 Stacked Cell Mechanics

Zolfagharian et al. [5] use a stack of cells to increase stroke. In the ideal series model used here:

$$d_{f,N} = N\,d_{f,1} \quad \text{(total stroke)} \tag{16}$$

$$d_{s,N} = N\,d_{s,1} \quad \text{(saddle position scales proportionally)} \tag{17}$$

If the cells are identical and snap sequentially, the peak force threshold is approximately unchanged: $F_{\text{push},N} \approx F_{\text{push},1}$.

The coefficient A for the stacked system relates to the per-cell coefficient:

$$A_N = \frac{A_1}{N^3} \tag{18}$$

which follows because the cubic force polynomial scales as $A_N\cdot(Nd_p)(Nd_p-Nd_s)(Nd_p-Nd_f) = N^3\,A_N\cdot d_p(d_p-d_s)(d_p-d_f) = F_{\text{push}}$.

The effective stiffnesses scale as:

$$k_{\text{eff},1,N} = A_N \cdot Nd_s \cdot Nd_f = \frac{A_1\, d_s\, d_f}{N} = \frac{k_{\text{eff},1,1}}{N} \tag{19}$$

Thus adding cells *reduces stiffness* and *lowers natural frequency*:

$$f_{n,1,N} = \frac{f_{n,1,1}}{\sqrt{N}} \tag{20}$$

This provides a design knob: choose N to target a specific natural frequency.

2.7 1-DOF Hook Snap-Through Model

The hook mass m is connected to the fixed robot body through the bistable mechanism. External vibration $F_{\text{exc}}(t)$ is transmitted through the robot structure. The equation of motion is:

$$m\ddot{d} + c\,\dot{d} + F_{\text{restore}}(d) = F_{\text{exc}}(t) \tag{21}$$

where $c = 2\zeta m\omega_{n,1}$. The excitation is a linear chirp (frequency-modulated sinusoid):

$$F_{\text{exc}}(t) = F_0 \sin\left[2\pi\left(f_{\text{lo}}\, t + \frac{f_{\text{hi}} - f_{\text{lo}}}{2t_f} t^2\right)\right] \tag{22}$$

where f_{lo}, f_{hi} are the chirp bounds [Hz] and t_f is the simulation duration [s]. As the instantaneous frequency passes through $f_{n,1}$, resonance amplifies the displacement until snap-through.

2.8 2-DOF Tuned Mass Damper Model

Following Zolfagharian et al. [5] (their Figure 4), the main structure mass m_1 is supported by spring k_{main} and driven harmonically. The TMD mass m_2 is attached to the same flexible ruler at distance L_2 from the fixed end, where the bistable mechanism repositions it.

The coupled equations of motion are:

$$m_1\ddot{x}_1 + c_{\text{main}}\dot{x}_1 + k_{\text{main}}x_1 + k_b(x_1 - x_2) + c_b(\dot{x}_1 - \dot{x}_2) = F_0 \sin(\omega t) \tag{23}$$

$$m_2\ddot{x}_2 - k_b(x_1 - x_2) - c_b(\dot{x}_1 - \dot{x}_2) = 0 \tag{24}$$

where k_b and c_b are the stiffness and damping of the bistable TMD connection, linearized at the current stable state.

2.8.1 Stiffness of the Flexible Ruler

The effective spring stiffness of a flexible cantilever beam (ruler) at distance L from the fixed support is [5]:

$$k = \frac{3EI_r}{L^3} \tag{25}$$

where E_r is the ruler's elastic modulus, I_r is its second moment of area. When the bistable mechanism shifts the TMD mass from L_1 to $L_2 = L_1 + d_f$:

$$k_{\text{TMD},2} = \frac{3E_rI_r}{L_2^3} = \frac{3E_rI_r}{(L_1 + d_f)^3} < k_{\text{TMD},1} \tag{26}$$

This reduction in TMD stiffness shifts the TMD natural frequency downward, changing the frequency at which it suppresses main-mass vibration.

2.8.2 Steady-State Frequency Response

In the frequency domain, the steady-state response of x_1 to $F_0 e^{i\omega t}$ is $X_1 e^{i\omega t}$. The impedance formulation gives:

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} = \begin{Bmatrix} F_0 \\ 0 \end{Bmatrix} \tag{27}$$

where

$$Z_{11} = (k_{\text{main}} + k_b - m_1\omega^2) + i\omega(c_{\text{main}} + c_b) \tag{28}$$

$$Z_{12} = Z_{21} = -(k_b + i\omega c_b) \tag{29}$$

$$Z_{22} = (k_b - m_2\omega^2) + i\omega c_b \tag{30}$$

Solving by Cramer's rule:

$$X_1 = \frac{F_0\, Z_{22}}{Z_{11}Z_{22} - Z_{12}^2} \tag{31}$$

The amplitude $|X_1(\omega)|$ vs. $\omega/(2\pi)$ [Hz] is plotted for three configurations: no TMD, TMD in State 1, TMD in State 2.

3 Base Excitation via Motor Torque Oscillation Through Shared Structure

3.1 Physical Setup

3.1.1 Mechanism component chain

The Zolfagharian Design 9 bistable mechanism consists of the following rigid bodies connected in series:

1. **Robot chassis / ground** — the fixed reference frame.
2. **Spring isolators** (k_s, c_s) — soft mounting elements (rubber mounts, vibration isolators) between the chassis and the plate assembly.
3. **Rigid plate assembly** (M_P) — one rigid body comprising the plate, the motor housing, and the *external blocks*. The external blocks are the side walls to which the bistable beam ends are pinned.
4. **Bistable curved beams** — the compliant Qiu cosine-profile beams spanning between the external blocks. These provide the nonlinear restoring force $F_{\text{restore}}(x_{\text{rel}})$ and potential energy $V(x_{\text{rel}})$.
5. **Hook** (m_h, also called the *reference mass*) — the deployable element attached at the beam apex. In State 1 the beam is arched and the hook is retracted; in State 2 the beam has snapped through and the hook is deployed.

The hook is connected to the plate assembly *only* through the bistable beams — there is no rigid link. When the plate assembly vibrates, the hook cannot follow instantaneously; the resulting **relative displacement** x_{rel} between the beam apex (hook) and the beam ends (external blocks on plate) is the quantity that drives snap-through.

```
ROBOT CHASSIS
      |
      |  k_s, c_s   <- spring isolators / rubber feet
      |
RIGID ASSEMBLY  (M_P = motor + plate + external blocks)
      |                  external blocks = beam end pinned supports
      |
    BEAM      BEAM    <- bistable curved beams (Qiu cosine profile)
      |          |
      +----+-----+
           |
         HOOK  (m_h = reference mass at beam apex)
```

Figure 1: Physical component chain for the Zolfagharian bistable deployment mechanism.

3.1.2 Two excitation configurations

Configuration 1 — Motor on plate assembly. The motor housing is bolted directly to the plate (part of M_P). The motor shaft drives an eccentric mass or proof mass whose reaction force enters the plate through the bearing. The plate assembly vibrates; the hook responds via base excitation through the beams.

$$\text{Motor} \;\rightarrow\; \text{Plate} + \text{External blocks} \;\rightarrow\; \text{Beams} \;\rightarrow\; \text{Hook}$$

Configuration 2 — Motor off plate, drives via bearing. The motor housing is *not* part of the plate assembly. The motor output shaft drives the plate+external-blocks assembly directly through a bearing coupling (lever arm, crank, or shaft). In this configuration the motor can *prescribe* the plate displacement:

$$x_P(t) = A\,\sin(\omega_{n,1}\,t)$$

The hook responds to this base excitation through the beams. The motor housing mass is excluded from M_P.

$$\text{Motor} \;\rightarrow\; \text{Bearing} \;\rightarrow\; \text{Plate} + \text{External blocks} \;\rightarrow\; \text{Beams} \;\rightarrow\; \text{Hook}$$

Configuration 2 avoids the tuned-mass-damper anti-resonance problem (see Section 3.10) because the motor enforces the plate amplitude regardless of the hook's dynamics.

3.1.3 Coordinates

Two coordinate systems are used throughout:

- $x_P(t)$: absolute displacement of the rigid plate assembly [m] (positive in the snap direction, i.e. from State 1 toward State 2)
- $x_H(t)$: absolute hook displacement [m]
- $x_{\mathrm{rel}}(t) \equiv x_H(t) - x_P(t)$: hook displacement *relative to the plate assembly* [m] — this is the quantity that enters the bistable restoring force and potential energy

Snap-through occurs when x_{rel} crosses the saddle point d_s with positive velocity.

3.2 Motor Torque Excitation Force

In torque/current control mode the RS05 applies a sinusoidal output torque $\tau_{\text{out}}(t)$. Through a coupling of moment arm r_{arm} [m] this becomes a linear force on the plate:

$$F_{\text{motor}}(t) = \frac{\tau_{\text{out}}}{r_{\text{arm}}} \sin(\omega_d t) \equiv F_0 \sin(\omega_d t) \tag{32}$$

where τ_{out} is the **measured output torque** read from the motor's torque–speed (T-N) curve at the instantaneous operating condition, and ω_d is the drive frequency set by the controller command. The amplitude F_0 is therefore *independent of frequency* — a key difference from a rotating-unbalance source whose amplitude grew as ω^2.

For a linear chirp sweep from f_{lo} to f_{hi} over total time t_f, the instantaneous frequency and accumulated phase are:

$$f_{\text{inst}}(t) = f_{\text{lo}} + \frac{f_{\text{hi}} - f_{\text{lo}}}{t_f}\, t \tag{33}$$

$$\varphi(t) = 2\pi \left[f_{\text{lo}}\, t + \frac{f_{\text{hi}} - f_{\text{lo}}}{2t_f}\, t^2 \right] \tag{34}$$

The torque-driven force during a chirp is:

$$F_{\text{motor}}(t) = F_0 \sin(\varphi(t)) \tag{35}$$

The amplitude envelope F_0 is constant throughout the sweep, so the force delivered at $f_{n,1}$ equals the force at any other frequency in the band — the mechanism receives full excitation amplitude at every point in the chirp.

3.3 Gearbox Effects on Vibration Transmission

Any gearbox in the drivetrain between the motor rotor and the bistable plate acts as a mechanical filter. The gear ratio n_g amplifies the rotor's reflected inertia and creates a low-pass cut-off frequency above which vibrations are attenuated before reaching the plate. Two architectures relevant to bistable excitation are treated below.

3.3.1 Planetary Gearboxes

A planetary gearbox achieves its ratio n_g through a sun–planet–ring arrangement. Typical ratios range from 3:1 to about 20:1 per stage; multi-stage units reach 100:1. QDD motors use a single low-ratio planetary stage ($n_g \lesssim 10$) specifically to preserve backdrivability and broadband force transmission.

Reflected rotor inertia. At the output shaft, the effective inertia is:

$$J_{\text{eff}} = J_{\text{rotor}}\, n_g^2 + J_{\text{gears}} + J_{\text{output}} \tag{36}$$

The n_g^2 factor dominates: even a modest 5:1 ratio makes the rotor inertia appear 25× larger at the output. Combined with the finite gear-mesh stiffness k_{gear}, this creates a mechanical low-pass cut-off frequency:

$$f_{\text{cut}} \approx \frac{1}{2\pi} \sqrt{\frac{k_{\text{gear}}}{J_{\text{eff}}}} \tag{37}$$

Vibrations at frequencies above f_{cut} are attenuated before reaching the plate; those well below f_{cut} pass through with little attenuation. A low gear ratio keeps f_{cut} high, so the motor remains substantially backdrivable and transmits a broad vibration band — a favorable property for this application.

Design rule: verify that $f_{n,1} < f_{\text{cut}}$ so that the bistable resonance lies in the pass-band of the actuator's mechanical filter. This check should use the actual mounted motor and linkage geometry, not the bare motor datasheet value.

Gear-mesh frequency. The planet-gear tooth contacts produce a secondary vibration at:

$$f_{\text{mesh}} = n_g\, f_{\text{rotor}}\, N_{\text{teeth}} \tag{38}$$

where N_{teeth} is the planet tooth count. If f_{mesh} happens to coincide with $f_{n,1}$, this harmonic can contribute beneficially to snap-through excitation without any additional control effort. A low single-stage ratio keeps f_{mesh} well above the bistable resonance band, avoiding unintended parametric excitation.

3.3.2 High-Ratio Straight-Tooth Gearboxes

Multi-stage spur or helical gearboxes achieve ratios of $n_g = 20{:}1$ to $n_g > 100{:}1$ by cascading two or more stages of spur (straight-tooth) or helical gears. They are inexpensive and compact relative to harmonic or cycloidal alternatives at the same ratio, but carry two significant penalties for bistable vibration excitation.

Reflected inertia penalty. Because n_g is large, the n_g^2 term in eq. (36) dominates overwhelmingly. The cut-off frequency f_{cut} drops accordingly:

$$f_{\text{cut}} \;\propto\; \frac{1}{n_g} \qquad (J_{\text{eff}} \approx J_{\text{rotor}}\, n_g^2) \tag{39}$$

At $n_g = 50$, a motor with $f_{\text{cut}} = 500\,\text{Hz}$ at $n_g = 1$ drops to roughly $10\,\text{Hz}$ — potentially *below* the bistable resonance. Vibrations above this cut-off are mechanically filtered; the plate never receives the high-frequency force content needed for resonant snap-through.

Backlash dead-band. Multi-stage spur gearboxes accumulate backlash θ_{bl} (typically 5–30 arc-min total) from each mesh. Through the coupling lever arm r this creates a linear dead-band:

$$\delta_{bl} = \theta_{bl}\, r \tag{40}$$

During each oscillation half-cycle the drive must traverse δ_{bl} before force transmission resumes. The effective plate amplitude is reduced to:

$$A_{\text{eff}} = A_{\text{plate}} - \delta_{bl} \tag{41}$$

For snap-through, the minimum plate amplitude from eq. (103) must be met *net of backlash*: $A_{\text{eff}} \geq A_{\min}$, i.e. $A_{\text{plate}} \geq 2\zeta_h\, d_s + \delta_{bl}$. If δ_{bl} is comparable to d_s, this constraint can become severe.

Gear-mesh noise. Multi-stage meshes generate harmonics at integer multiples of f_{mesh} (eq. (38)). If any harmonic coincides with $f_{n,1}$, it can excite unintended snap-through; this is harder to avoid in a multi-stage box than in a single-stage planetary.

Design rule. High-ratio straight-tooth gearboxes are suitable for bistable excitation only when $f_{n,1} \ll f_{\text{cut}}(n_g)$, the accumulated backlash $\delta_{bl} \ll d_s$, and the drive frequency is well below the first gear-mesh harmonic. In practice, these conditions are difficult to satisfy simultaneously at gear ratios above $\sim 20{:}1$; a low-ratio QDD or cycloidal stage is strongly preferred.

3.4 Two-Degree-of-Freedom Equations of Motion

The plate is modelled as a rigid body on a linear elastic support (stiffness k_s, damping c_s). The hook mass m_h is connected to the plate through the bistable element (restoring force $F_{\text{restore}}(x_{\text{rel}})$, eq. (9)) and a viscous damper c_h.

Plate equation of motion:

$$M_P \ddot{x}_P = F_{\text{motor}}(t) \underbrace{- k_s x_P - c_s \dot{x}_P}_{\text{support restraint}} \underbrace{- F_{\text{restore}}(x_{\text{rel}}) + c_h \dot{x}_{\text{rel}}}_{\text{reaction from hook}} \tag{42}$$

Hook equation of motion (absolute frame):

$$m_h \ddot{x}_H = F_{\text{restore}}(x_{\text{rel}}) - c_h \dot{x}_{\text{rel}} \tag{43}$$

The sign convention in eq. (42) reflects Newton's third law: the bistable spring pulls the plate with force $-F_{\text{restore}}$ (equal and opposite to what it exerts on the hook), and the dashpot exerts $+c_h \dot{x}_{\text{rel}}$ on the plate.

3.4.1 Relative-Coordinate Formulation

Subtracting eq. (42) (divided by M_P) from eq. (43) (divided by m_h):

$$\ddot{x}_{\text{rel}} = \ddot{x}_H - \ddot{x}_P = \underbrace{\frac{F_{\text{restore}} - c_h \dot{x}_{\text{rel}}}{m_h}}_{\ddot{x}_H} - \underbrace{\frac{F_{\text{motor}} - k_s x_P - c_s \dot{x}_P - F_{\text{restore}} + c_h \dot{x}_{\text{rel}}}{M_P}}_{\ddot{x}_P} \tag{44}$$

This is the governing equation for the bistable snap. The full state vector is $\mathbf{y} = [x_P,\, \dot{x}_P,\, x_{\text{rel}},\, \dot{x}_{\text{rel}}]^\top$.

The support parameters are defined from the plate natural frequency f_P:

$$k_s = M_P\,(2\pi f_P)^2 \tag{45}$$

$$c_s = 2\,\zeta_P\, M_P\,(2\pi f_P) \tag{46}$$

where ζ_P is the plate structural damping ratio.

3.5 Plate Transfer Function (Decoupled Analysis)

For small x_{rel} (early in the simulation, well before snap), the back-reaction of the bistable on the plate is negligible relative to the dominant terms F_{motor}, $k_s x_P$, and $c_s \dot{x}_P$. Under this approximation, the plate behaves as an independent SDOF oscillator:

$$M_P \ddot{x}_P + c_s \dot{x}_P + k_s x_P \approx F_{\text{motor}}(t) \tag{47}$$

Assuming harmonic excitation at frequency ω with constant amplitude F_0, the steady-state plate displacement amplitude is:

$$\left|X_P(\omega)\right| = \frac{F_0}{\sqrt{\left(k_s - M_P \omega^2\right)^2 + \left(c_s\,\omega\right)^2}} \tag{48}$$

Substituting eqs. (45) and (46) and defining the frequency ratio $r_P = \omega/\omega_P$ where $\omega_P = 2\pi f_P$:

$$\left|X_P(\omega)\right| = \frac{F_0/(M_P \omega_P^2)}{\sqrt{\left(1 - r_P^2\right)^2 + \left(2\zeta_P r_P\right)^2}} = \frac{F_0/k_s}{\sqrt{(1 - r_P^2)^2 + (2\zeta_P r_P)^2}} \tag{49}$$

This is the standard frequency-response function of a driven SDOF oscillator. The amplitude peaks sharply at $r_P = 1$ (plate resonance, $\omega = \omega_P$), with the peak width governed by ζ_P.

3.6 Effective Excitation Force on the Bistable Hook

In the relative-coordinate frame, the plate's acceleration acts as an inertial (fictitious) force on the hook. The effective force amplitude at frequency ω is:

$$F_{\text{eff}}(\omega) = m_h\,\omega^2\,\big|X_P(\omega)\big| \tag{50}$$

Substituting eq. (48):

$$\boxed{F_{\text{eff}}(\omega) = \frac{m_h\,F_0\,\omega^2}{\sqrt{\big(k_s - M_P\omega^2\big)^2 + \big(c_s\,\omega\big)^2}}} \tag{51}$$

This is the force that the base excitation delivers to the bistable hook at frequency ω. Several observations follow directly:

1. The numerator scales as ω^2 (not ω^4 as in a rotating-unbalance source): the force still grows with frequency, but more gradually. At low frequencies the inertial coupling is weak; at high frequencies the hook is increasingly driven by the plate's acceleration.

2. The denominator contains the plate resonance denominator. When ω approaches ω_P, the denominator passes through a minimum, and F_{eff} passes through a maximum.

3. The maximum of F_{eff} occurs near (but not exactly at) ω_P. If ω_P is designed to coincide with $\omega_{n,1}$, the maximum excitation force is delivered precisely when the bistable hook is at its resonance — a double-resonance condition that amplifies the effective force by $Q_P \cdot Q_h$ relative to the quasi-static case.

3.7 General Snap-Through Condition

From eq. (99) (Optimization section), the minimum force required for snap-through at resonance $\omega = \omega_{n,1}$ is:

$$F_{0,\min} = 2\zeta_h\,k_{\text{eff},1}\,d_s = 2\zeta_h\,m_h\,\omega_{n,1}^2\,d_s \tag{52}$$

Setting $F_{\text{eff}}(\omega_{n,1}) \geq F_{0,\min}$ and substituting eq. (51):

$$\frac{m_h\,F_0\,\omega_{n,1}^2}{\sqrt{(k_s - M_P\omega_{n,1}^2)^2 + (c_s\,\omega_{n,1})^2}} \geq 2\zeta_h\,m_h\,\omega_{n,1}^2\,d_s \tag{53}$$

Dividing both sides by $m_h\,\omega_{n,1}^2$ (positive):

$$\frac{F_0}{\sqrt{(k_s - M_P\omega_{n,1}^2)^2 + (c_s\,\omega_{n,1})^2}} \geq 2\zeta_h\,d_s \tag{54}$$

Rearranging, the **general snap condition** on the motor force amplitude is:

$$\boxed{F_0 \geq 2\zeta_h\,d_s\,\sqrt{(k_s - M_P\omega_{n,1}^2)^2 + (c_s\,\omega_{n,1})^2}} \tag{55}$$

The right-hand side is entirely determined by known system parameters. Notably, m_h has cancelled: the snap condition is *independent of hook mass*. However, $\omega_{n,1}$ itself depends on m_h through eq. (13), so the dependence is implicit.

3.8 Case 1: Tuned Plate ($f_P = f_{n,1}$)

3.8.1 Derivation

Set the plate support natural frequency equal to the bistable natural frequency:

$$f_P = f_{n,1} \quad \Longleftrightarrow \quad \omega_P = \omega_{n,1} \quad \Longleftrightarrow \quad k_s = M_P\,\omega_{n,1}^2 \tag{56}$$

Substituting into eq. (54). The term $k_s - M_P\omega_{n,1}^2$ vanishes exactly, leaving only the damping term in the denominator:

$$\sqrt{(M_P\omega_{n,1}^2 - M_P\omega_{n,1}^2)^2 + (c_s\,\omega_{n,1})^2} = c_s\,\omega_{n,1} \tag{57}$$

Substituting $c_s = 2\zeta_P M_P \omega_P = 2\zeta_P M_P \omega_{n,1}$:

$$c_s\,\omega_{n,1} = 2\zeta_P\,M_P\,\omega_{n,1}^2 \tag{58}$$

The snap condition eq. (54) becomes:

$$\frac{F_0}{2\zeta_P\,M_P\,\omega_{n,1}^2} \geq 2\zeta_h\,d_s \tag{59}$$

$$\boxed{F_0 \;\geq\; 4\,\zeta_h\,\zeta_P\,M_P\,\omega_{n,1}^2\,d_s \qquad \text{(Case 1: tuned plate)}} \tag{60}$$

3.8.2 Physical Interpretation

Two points are useful for design:

1. **Scales with frequency.** Unlike a rotating-unbalance source, $F_0 \propto \omega_{n,1}^2$: a higher bistable resonance requires more actuator force. The available force should be verified against this threshold after any change to geometry or mass, using the actuator's rated output at the chosen coupling arm length.

2. **Doubly discounted by damping.** The factor $4\zeta_h\zeta_P$ means the threshold is $Q_P = 1/(2\zeta_P)$ times smaller than the off-resonance quasi-static baseline $2\zeta_h M_P \omega_{n,1}^2 d_s$. This Q_P factor is the mechanical quality factor of the plate resonance.

   Physical explanation: at plate resonance the plate amplitude is amplified by Q_P relative to its quasi-static response; the hook then experiences Q_P times more inertial excitation, so Q_P times less actuator force is required.

3.8.3 Plate Amplitude at Snap Threshold

At the tuned threshold, from eq. (48) with the denominator reduced to $c_s\omega_{n,1}$:

$$|X_P|_{\text{tuned}} = \frac{F_0}{c_s\,\omega_{n,1}} = \frac{4\zeta_h\zeta_P M_P \omega_{n,1}^2 d_s}{2\zeta_P M_P \omega_{n,1}^2} = 2\zeta_h\,d_s \tag{61}$$

The plate displacement at threshold is $2\zeta_h d_s$. The hook displacement, amplified by $Q_h = 1/(2\zeta_h)$ at bistable resonance, reaches d_s:

$$|x_{\text{rel}}|_{\text{at snap}} = Q_h\,\left|X_P\right| = \frac{2\zeta_h\,d_s}{2\zeta_h} = d_s \tag{62}$$

as expected. The two resonances (plate at f_P, hook at $f_{n,1}$) in series provide amplification $Q_P \cdot Q_h = 1/(4\zeta_P\zeta_h)$ relative to the quasi-static case.

3.9 Case 2: Stiffness-Controlled Plate ($f_P > f_{n,1}$)

3.9.1 Regime Identification

When the plate support is stiffer than needed for resonance, $f_P > f_{n,1}$, the plate operates in the *stiffness-controlled* regime at the bistable's resonant frequency. Define the detuning ratio:

$$n \equiv \frac{f_P}{f_{n,1}} = \frac{\omega_P}{\omega_{n,1}} > 1 \tag{63}$$

In this regime $k_s > M_P\omega_{n,1}^2$, so the term $k_s - M_P\omega_{n,1}^2 = M_P(\omega_P^2 - \omega_{n,1}^2) > 0$ is the dominant contribution to the denominator of eq. (54).

3.9.2 Derivation

Substituting $k_s = M_P\omega_P^2 = M_P n^2\omega_{n,1}^2$ and $c_s\omega_{n,1} = 2\zeta_P M_P\omega_P\omega_{n,1} = 2\zeta_P M_P n\omega_{n,1}^2$ into the denominator of eq. (55):

$$\begin{aligned}&\sqrt{(k_s - M_P\omega_{n,1}^2)^2 + (c_s\omega_{n,1})^2}\\ &= \sqrt{M_P^2\omega_{n,1}^4(n^2-1)^2 + M_P^2\omega_{n,1}^4(2\zeta_P n)^2}\\ &= M_P\,\omega_{n,1}^2\sqrt{(n^2-1)^2 + (2\zeta_P n)^2}\end{aligned} \tag{64}$$

Substituting into eq. (55):

$$\boxed{F_0 \;\geq\; 2\zeta_h\, M_P\, \omega_{n,1}^2\, d_s\, \sqrt{(n^2-1)^2 + (2\zeta_P n)^2} \qquad \text{(Case 2: } n = f_P/f_{n,1} > 1)} \tag{65}$$

For light structural damping ($\zeta_P \ll 1$, so $2\zeta_P n \ll n^2 - 1$):

$$F_0 \;\approx\; 2\zeta_h\, M_P\, \omega_{n,1}^2\, d_s\, (n^2-1) \qquad (n > 1,\; \zeta_P \ll 1) \tag{66}$$

The threshold grows as $(n^2 - 1)$, which is approximately n^2 for large n. A plate resonance twice as high as the bistable resonance ($n = 2$) requires approximately $3\times$ the off-resonance inertia floor, and $75\times$ the tuned threshold.

3.9.3 Verification: Case 2 Reduces to Case 1 at $n = 1$

Setting $n = 1$ in eq. (65):

$$F_0 \geq 2\zeta_h\, M_P\, \omega_{n,1}^2\, d_s\, \sqrt{0 + (2\zeta_P)^2} = 4\zeta_h\zeta_P\, M_P\, \omega_{n,1}^2\, d_s \tag{67}$$

which recovers eq. (60).

3.10 Comparison of Cases and Design Guidance

3.10.1 Closed-Form Comparison

Define the off-resonance baseline $T_0 = 2\zeta_h M_P\omega_{n,1}^2 d_s$. Then:

$$\text{Case 1 (tuned):}\quad F_0 \geq 4\zeta_h\zeta_P\, M_P\, \omega_{n,1}^2\, d_s = 2\zeta_P\, T_0 \tag{68}$$

$$\text{Case 2 (stiffness-controlled):}\quad F_0 \geq T_0\, \sqrt{(n^2-1)^2 + (2\zeta_P n)^2} \tag{69}$$

The ratio of Case 2 to Case 1 thresholds is unchanged from the rotating-unbalance analysis:

$$\frac{\text{Case 2 threshold}}{\text{Case 1 threshold}} = \frac{\sqrt{(n^2-1)^2 + (2\zeta_P n)^2}}{2\zeta_P} \approx \frac{n^2-1}{2\zeta_P} \qquad (n > 1,\ \zeta_P \ll 1) \tag{70}$$

For $\zeta_P = 0.02$ and $n = 2$: ratio $\approx (4-1)/(0.04) = 75$. For $n = 1.5$: ratio $\approx (2.25-1)/0.04 = 31$. For $n = 1.1$: ratio $\approx (1.21-1)/0.04 = 5.3$.

Even a modest 10% detuning ($n = 1.1$) increases the required motor force by more than 5× compared to the tuned case.

3.10.2 Design Guidelines

The analysis above yields three actionable design rules:

1. **Tune the plate.** Set the mounting plate's natural frequency to match the bistable mechanism's natural frequency by choosing the stiffness of the isolators or mounting brackets:

$$k_s = M_P \,(2\pi f_{n,1})^2 \tag{71}$$

   Even an approximate match ($n \leq 1.1$) strongly reduces the required force compared with a stiff, uncontrolled plate. Note that M_P here is the mass of the *plate and bistable assembly only* — the motor body should be mounted separately to the chassis so that its mass does not add to M_P.

2. **Minimize plate damping ζ_P.** The tuned threshold is $4\zeta_h \zeta_P M_P \omega_{n,1}^2 d_s \propto \zeta_P$. Using low-loss-factor rubber isolators is preferred, provided the plate does not become too loose or poorly constrained for the robot packaging.

3. **Verify the gearbox pass-band.** Any gearbox in the drivetrain attenuates vibrations above f_{cut} (eq. (37)). Confirm that $f_{n,1} < f_{\text{cut}}$ so that the actuator force is not filtered before it reaches the bistable plate. High-ratio straight-tooth gearboxes are particularly susceptible (see Section 3.3.2); a low-ratio planetary or QDD stage is preferred. Use final mounted geometry rather than the bare datasheet value for this check.

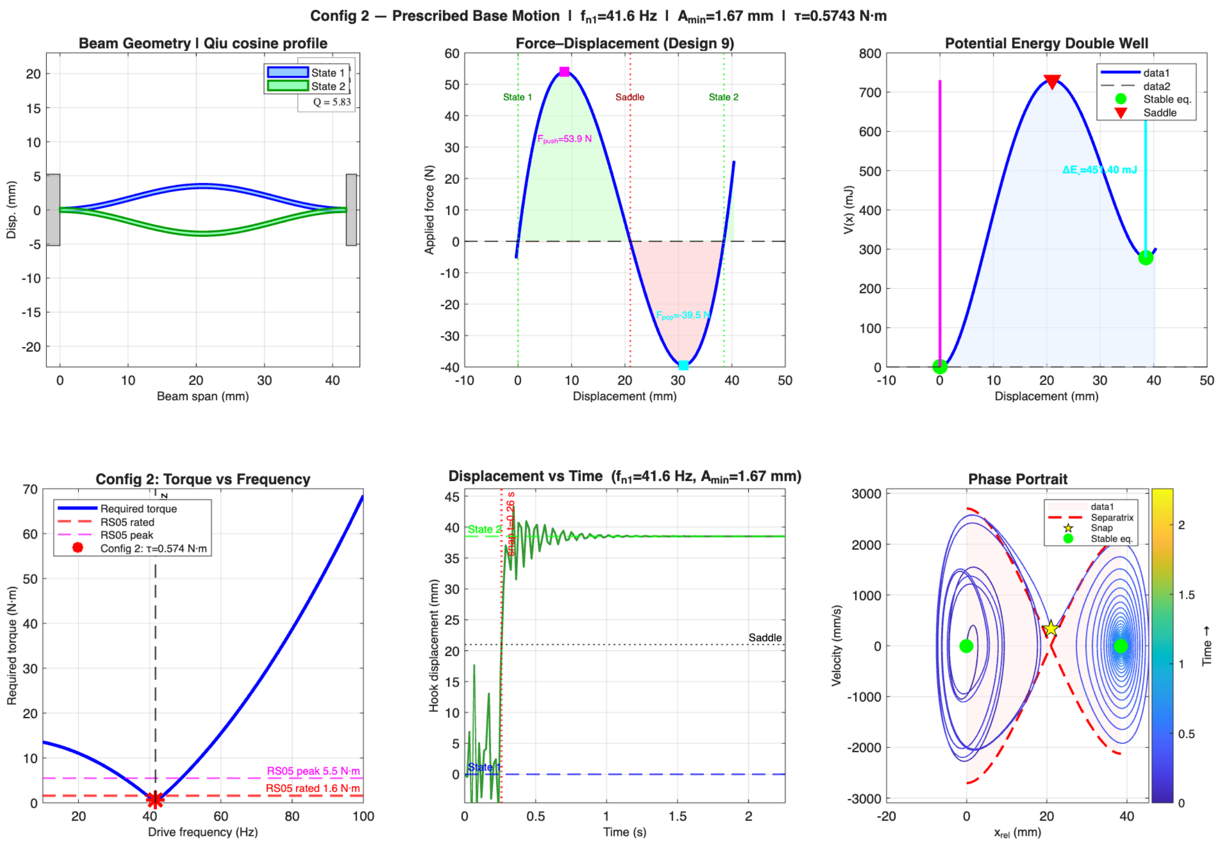


Figure 2: Time-domain snap-through simulation for Configuration 2 (prescribed base motion). The plate displacement $x_P(t)$ and relative hook displacement $x_{\mathrm{rel}}(t)$ are plotted; the hook crosses the saddle d_s and latches into State 2 without further motor input.

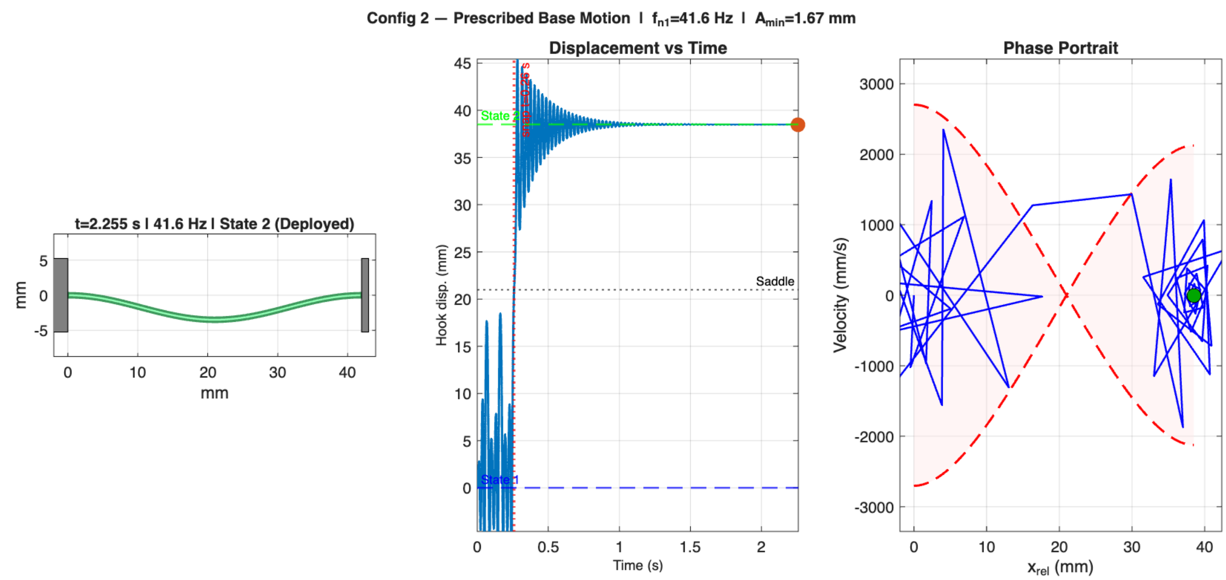


Figure 3: Animation snapshot of the Configuration 2 bistable excitation (prescribed base motion). The plate oscillates at $f_{n,1}$ and the hook builds resonant amplitude until snap-through occurs.

3.11 Motor Mounted on the Bistable Plate: Coupling Model and Assumptions

This subsection addresses the specific physical configuration assumed throughout Section 3: the motor body is bolted directly to the same rigid plate that carries the bistable mechanism. It tracks the force path from an eccentric mass to the plate, shows why shaft angle is not a state variable in the hook dynamics, and states the assumptions explicitly.

3.11.1 Force Transmission Chain

When the motor shaft rotates, the eccentric mass m_e at radius e generates a centrifugal force. Before this force reaches the plate it passes through three mechanical interfaces, each with its own compliance:

1. **Shaft** (stiffness k_{sh}, damping c_{sh}): The shaft transmits the eccentric mass force to the inner bearing race. A rigid shaft ($k_{\text{sh}} \to \infty$) transmits 100%; a flexible shaft at speeds near its *critical speed* $\omega_{\text{crit}} = \sqrt{k_{\text{sh}}/m_e}$ can amplify or attenuate the transmitted force.

2. **Bearing** (stiffness k_b, damping c_b): The rolling-element bearing connects inner race to outer race (motor housing). The Hertzian contact stiffness of a typical deep-groove ball bearing is $k_b \sim 10^7$–10^8 N/m, far above the frequencies of interest here, so bearing deflection is negligible.

3. **Motor mount bolts and plate**: The motor housing is assumed rigidly bolted to the plate. Any compliance in this joint (bolt stretch, washer deformation) is folded into the effective bearing stiffness.

The *bearing transmission function* $T_b(\omega)$ quantifies how much of the raw centrifugal force $m_e e\omega^2$ actually reaches the plate at frequency ω:

$$T_b(\omega) = \frac{k_b + i\, c_b\, \omega}{\left(k_b - m_e\, \omega^2\right) + i\, c_b\, \omega} \tag{72}$$

The force entering the plate is therefore:

$$F_{\text{plate}}(\omega) = T_b(\omega) \cdot m_e\, e\, \omega^2 \tag{73}$$

Rigid-bearing limit. When $k_b \gg m_e\, \omega^2$ (bearing far from its own resonance):

$$T_b(\omega) \;\to\; 1 \qquad (k_b \to \infty) \tag{74}$$

All force transfers to the plate. Rolling-element bearings typically have Hertzian contact stiffnesses $k_b \sim 10^7$–10^8 N/m. As long as $m_e\, \omega^2 \ll k_b$ — satisfied at any bistable resonance frequency well below the bearing's own resonance — $T_b \approx 1$ and the full centrifugal force reaches the plate, justifying $F_{\text{motor}} = m_e\, e\, \omega^2 \, \sin(\omega t)$.

3.11.2 Single-Axis Projection of the 2-D Rotating Force

The centrifugal force is a vector that rotates in the plane of the motor shaft:

$$\mathbf{F}_{\text{cent}}(t) = m_e\, e\, \omega^2 \left[\cos(\omega t)\, \hat{x} + \sin(\omega t)\, \hat{y}\right] \tag{75}$$

Only the component aligned with the bistable stroke direction (take $\hat{y}$ as the snap axis) does useful work on the bistable:

$$F_{\text{snap}}(t) = \mathbf{F}_{\text{cent}}(t) \cdot \hat{y} = m_e\, e\, \omega^2 \, \sin(\omega t) \tag{76}$$

The orthogonal component $m_e e\omega^2 \cos(\omega t)$ loads the plate transversely; it is transmitted to the plate support as a lateral force but does not couple into the bistable snap direction and is neglected in the 1-DOF model.

Practical implication. Aligning the motor shaft so that the eccentricity plane contains the snap direction maximises the effective force. Misalignment by angle α reduces F_{snap} by $\cos\alpha$.

3.11.3 Motor Shaft Angle Does Not Appear in the Hook Dynamics

A key property of the rotating-unbalance architecture is that the *motor shaft angle* $\theta(t) = \omega t + \theta_0$ does not appear in the bistable hook's equation of motion — only the resulting plate acceleration does.

Proof. The hook EOM in relative coordinates is (eq. (44)):

$$m_h\, \ddot{x}_{\text{rel}} = F_{\text{restore}}(x_{\text{rel}}) - c_h\, \dot{x}_{\text{rel}} - m_h\, \ddot{x}_P(t) \tag{77}$$

The right-hand side contains $\ddot{x}_P(t)$ — the plate acceleration. This quantity depends on $F_{\text{snap}}(t) = m_e e\omega^2 \sin(\omega t)$ through the plate dynamics, but the shaft angle θ itself never appears. Whether the motor has rotated 100 turns or 1000 turns is irrelevant; only the instantaneous force amplitude and frequency matter.

Contrast with direct drive. In a crank-slider or lever-arm direct-drive arrangement, the motor shaft angle θ sets the hook position geometrically: $x_{\text{hook}} = f(\theta)$. A change in θ immediately moves the hook. With rotating unbalance there is no such kinematic constraint: the hook position is determined entirely by the energy balance between the inertial excitation and the bistable restoring force. The motor can spin freely at constant speed without tracking or position control.

3.11.4 Summary of Model Assumptions

Table 1: Assumptions in the shared-plate rotating-unbalance model and their validity for the Rolly deployment mechanism.

Assumption	**Mathematical form**	**Validity / break-down**
Rigid bearing	$T_b(\omega) = 1$	Valid: $k_b \gg m_e\omega^2$ at 19.7 Hz
Rigid motor shaft	$\omega_{\text{crit}} \gg \omega$	Valid if shaft critical speed $\gg$ 19.7 Hz
Constant motor speed	$\dot{\omega} = 0$	Valid if motor torque bandwidth $\gg f_{n,1}$
Single-axis force	$F = m_e e\omega^2 \sin(\omega t)$	Loses factor $\cos\alpha$ if shaft misaligned
Rigid motor-to-plate joint	$k_{\text{joint}} \to \infty$	Valid for bolted flange; invalid for press-fit
Plate as rigid body	No plate bending modes	Valid if plate structural modes $\gg f_{n,1}$
Linear plate support	k_s, c_s constant	Invalid near plate support nonlinearity

3.12 Motor Housing on the Plate with a Proof Mass on the Output Shaft

This subsection treats a distinct configuration: the motor housing is bolted to the bistable plate as before, but instead of a continuously spinning eccentric mass, the motor output shaft drives a *proof mass* M_{pm} that **oscillates linearly** along the snap axis relative to the plate. This is sometimes called a *reaction-mass actuator* or *proof-mass actuator* and is used when independent control of force amplitude and frequency is required.

3.12.1 Equations of Motion

Define coordinates:

- $x_P(t)$: absolute plate displacement (positive in snap direction)
- $\delta(t) = x_{pm}(t) - x_P(t)$: proof mass displacement *relative* to plate
- $x_{\text{rel}}(t)$: hook displacement relative to plate (unchanged from eq. (44))

The motor applies an internal force $F_{\text{act}}(t)$ between the plate (housing) and the proof mass. By Newton's third law the plate receives $-F_{\text{act}}$ and the proof mass receives $+F_{\text{act}}$.

Proof mass (absolute frame):

$$M_{pm}\,\ddot{x}_{pm} = M_{pm}\,(\ddot{x}_P + \ddot{\delta}) = F_{\text{act}}(t) \tag{78}$$

Plate (absolute frame):

$$M_P\,\ddot{x}_P = -F_{\text{act}}(t) - k_s\,x_P - c_s\,\dot{x}_P - F_{\text{restore}}(x_{\text{rel}}) + c_h\,\dot{x}_{\text{rel}} \tag{79}$$

Adding eqs. (78) and (79) to eliminate F_{act}:

$$\big(M_P + M_{pm}\big)\ddot{x}_P + M_{pm}\,\ddot{\delta} = -k_s\,x_P - c_s\,\dot{x}_P - F_{\text{restore}}(x_{\text{rel}}) + c_h\,\dot{x}_{\text{rel}} \tag{80}$$

The key observation is that F_{act} has been eliminated. The plate equation is now driven entirely by the proof mass *relative acceleration* $\ddot{\delta}$.

3.12.2 Prescribed Harmonic Proof-Mass Motion

If the motor drives the proof mass to oscillate at amplitude Δ and frequency ω:

$$\delta(t) = \Delta\,\sin(\omega t) \qquad \Longrightarrow \qquad \ddot{\delta}(t) = -\Delta\,\omega^2\,\sin(\omega t) \tag{81}$$

Substituting into eq. (80):

$$\big(M_P + M_{pm}\big)\ddot{x}_P = \underbrace{M_{pm}\,\Delta\,\omega^2\,\sin(\omega t)}_{\text{effective driving force}} - k_s\,x_P - c_s\,\dot{x}_P - F_{\text{restore}}(x_{\text{rel}}) + c_h\,\dot{x}_{\text{rel}} \tag{82}$$

This has *exactly the same form* as the rotating-unbalance plate equation (eq. (42)), with two substitutions:

$$m_e\,e \;\longrightarrow\; M_{pm}\,\Delta \quad \text{(force product)} \tag{83}$$

$$M_P \;\longrightarrow\; M_P + M_{pm} \quad \text{(effective plate mass)} \tag{84}$$

All results from Sections 3.5–3.10 carry over with these substitutions.

3.12.3 Snap Condition for the Proof-Mass Configuration

Applying the same derivation as section 3.8 with substitutions eqs. (83) and (84):
Tuned plate ($f_P = f_{n,1}$):

$$\boxed{M_{pm}\,\Delta \;\geq\; 4\,\zeta_h\,\zeta_P\,(M_P + M_{pm})\,d_s} \tag{85}$$

Defining the proof-mass ratio $\mu = M_{pm}/M_P$ and rearranging:

$$\Delta \;\geq\; 4\,\zeta_h\,\zeta_P\,M_P\,d_s\,\frac{1+\mu}{\mu} = \frac{4\,\zeta_h\,\zeta_P\,M_P\,d_s}{\mu} + 4\,\zeta_h\,\zeta_P\,d_s \tag{86}$$

As $\mu \to \infty$ (proof mass much heavier than plate), the required stroke $\Delta \to 4\zeta_h\zeta_P d_s$, independent of M_P. As $\mu \to 0$ (very light proof mass), $\Delta \to \infty$: a tiny mass must oscillate with enormous stroke to produce enough force.

3.12.4 Comparison with Rotating Unbalance

Table 2: Proof-mass oscillator vs. rotating unbalance: key differences for the same bistable plate.

Property	**Rotating unbalance**	**Proof-mass oscillator**
Force amplitude	$m_e\, e\, \omega^2$ (fixed by geometry)	$M_{pm}\, \Delta\, \omega^2$ (Δ controllable)
Force–frequency	Always grows as ω^2	Grows as ω^2 unless $\Delta(\omega)$ is adjusted
Effective plate mass	M_P	$M_P + M_{pm}$
Motor direction	Continuous rotation (one direction)	Reciprocating (reverses each half-cycle)
Snap product	$m_e e \geq 4\zeta_h \zeta_P M_P d_s$	$M_{pm}\Delta \geq 4\zeta_h \zeta_P (M_P + M_{pm}) d_s$
Shaft angle in EOM?	No	No (same decoupling argument)

The proof-mass oscillator is strictly more flexible than the rotating unbalance because Δ can be varied in real time via motor current. The cost is that the motor must reverse direction each cycle rather than spinning continuously, and the heavier effective plate mass $(M_P + M_{pm})$ raises the snap threshold.

3.12.5 Proof-Mass Resonance

If the shaft connecting motor to proof mass has compliance k_{pm}, the proof mass forms a secondary resonance:

$$\omega_{pm} = \sqrt{\frac{k_{pm}}{M_{pm}}} \tag{87}$$

Driving near ω_{pm} amplifies Δ by $Q_{pm} = 1/(2\zeta_{pm})$, reducing the required motor force by the same factor. This creates a *triple resonance* path: plate resonance at ω_P, bistable hook resonance at $\omega_{n,1}$, and proof-mass resonance at ω_{pm}. Tuning all three to the same frequency gives amplification $Q_P \cdot Q_h \cdot Q_{pm}$ but also narrows the bandwidth and makes the system sensitive to parameter variation.

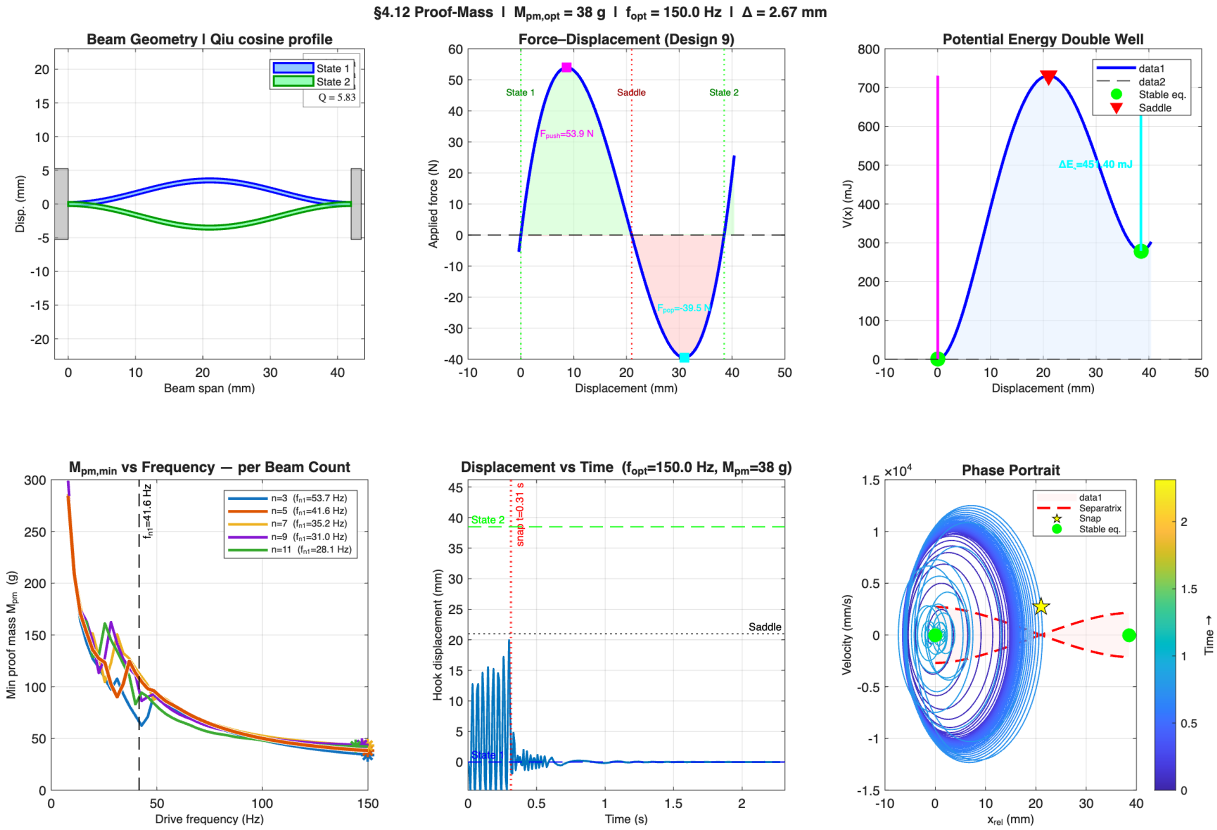


Figure 4: Proof-mass actuation analysis: minimum proof mass $M_{pm,\min}$ and corresponding snap-through drive frequency as a function of design parameters (N cells, hook mass m_h).

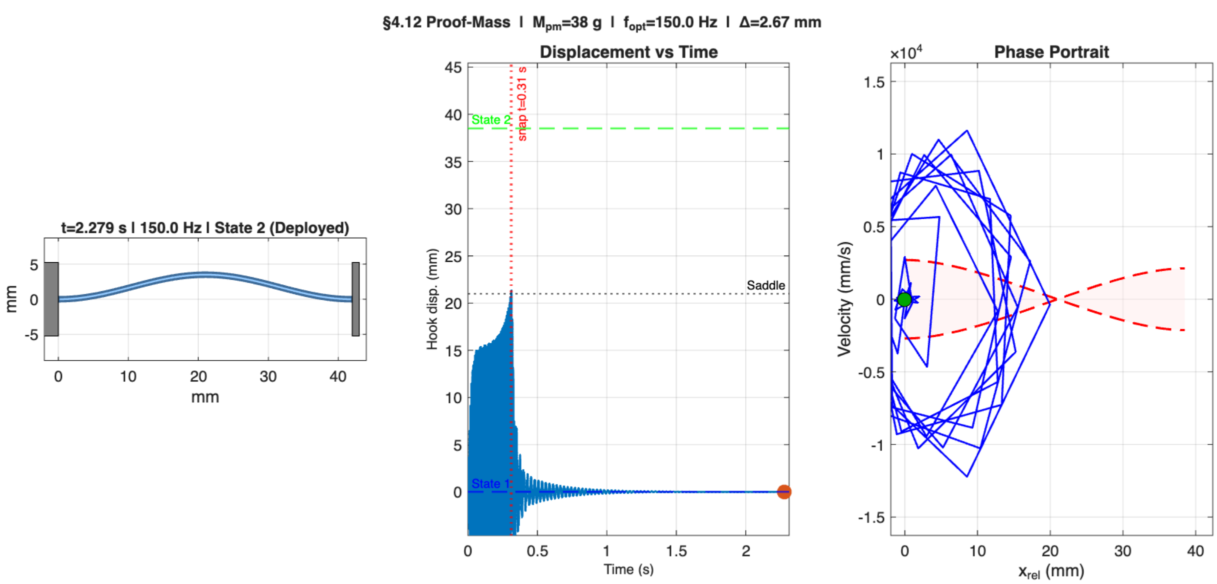


Figure 5: Animation snapshot for the Section 3.12 proof-mass actuator at the optimal proof-mass parameter. The proof mass oscillates on the motor output shaft, driving the plate and hook into snap-through.

4 RS05 Motor — Specifications and CAN Control for Bistable Excitation

This section documents the RobStride RS05 quasi-direct-drive motor specifications and CAN communication protocol as they relate to commanding sinusoidal vibration for bistable snap-through. All values are taken directly from the RS05 User Manual (Rev. 260428) [7].

4.1 Motor Specifications

Table 3: RS05 key parameters relevant to bistable excitation.

Parameter	**Value**	**Notes**
Rated torque	1.6 N·m	At 100 RPM, 70×70 mm heat sink
Peak torque	5.5 N·m	Short duration only (see overload table)
No-load speed	480 RPM ±10%	Output shaft; ≈ 50.3 rad/s = 8 Hz
Torque constant K_t	0.94 N·m/A_{rms}	Converts Iq current to torque
Deceleration ratio	7.75:1	Planetary gearbox
Drive mode	FOC	Field-oriented control
Max phase current	11 A_{pk}	Gives peak torque $\approx 11 \times 0.94/\sqrt{2} \approx$ 7.3 N·m (limited to 5.5)
Rated phase current	2.4 A_{pk}	Continuous
CAN bus rate	1 Mbps	CAN 2.0B, 29-bit extended frame
Encoder	14-bit absolute	$\pm 4\pi$ range (Type 2 update firmware)
Velocity feedback	±50 rad/s	Mapped [0–65535] in Type 2 frame
Torque feedback	±5.5 N·m	Mapped [0–65535] in Type 2 frame

4.2 Speed Limit and Implications for Bistable Excitation

The 480 RPM rating is a **shaft angular velocity limit** ($\omega_{\max} = 50.3$ rad/s), not a frequency limit. Its effect depends on whether the shaft undergoes *continuous rotation* or *oscillating (reciprocating) motion*.

4.2.1 Rotating Unbalance — Continuous Rotation (Infeasible)

A rotating unbalance requires the shaft to spin continuously at the bistable natural frequency $f_{n,1}$. To do so:

$$n_{\text{required}} = f_{n,1} \times 60 \tag{88}$$

For $f_{n,1} \approx 20$ Hz (current design parameters): $n = 1200$ RPM, which is 2.5× the 480 RPM limit. A rotating unbalance is therefore infeasible on the RS05 at the bistable resonance frequency.

4.2.2 Oscillating Excitation — Velocity, Not Frequency, Is the Constraint

For any *sinusoidal oscillation* (proof-mass or prescribed base motion) at frequency f and angular shaft amplitude Θ (rad):

$$v_{\text{peak}} = \Theta \cdot 2\pi f \ \leq \ \omega_{\text{max}} = 50.3 \text{ rad/s} \tag{89}$$

The constraint limits the *amplitude*, not the frequency. At any frequency the motor can oscillate with amplitude:

$$\Theta_{\text{max}}(f) = \frac{\omega_{\text{max}}}{2\pi f} \tag{90}$$

The corresponding maximum linear stroke through lever arm r is:

$$\delta_{\text{max}}(f) = \Theta_{\text{max}}(f) \cdot r = \frac{\omega_{\text{max}}\, r}{2\pi f} \tag{91}$$

For snap-through the stroke must reach the saddle: $\delta_{\text{max}} \geq d_s$. This gives a minimum lever arm for a given drive frequency:

$$r_{\text{min}}(f) = \frac{2\pi f\, d_s}{\omega_{\text{max}}} \tag{92}$$

Numerical example (current design). With $m_h = 0.200\,\text{kg}$, $N = 5$ cells, $d_{f,1} = 8.25\,\text{mm}$: $d_s \approx 22.5\,\text{mm}$, $f_{n,1} \approx 19.7\,\text{Hz}$.

$$\Theta_{\text{max}}(f_{n,1}) = 50.3/(2\pi \times 19.7) = 0.406 \text{ rad} = 23.3^\circ \tag{93}$$

$$\delta_{\text{max}}(f_{n,1}) = 0.406 \times 0.050 = 20.3 \text{ mm} \quad (r = 50 \text{ mm}) \tag{94}$$

Since $d_s = 22.5\,\text{mm} > 20.3\,\text{mm}$, a 50 mm lever arm is marginally insufficient. The minimum lever arm required is:

$$r_{\text{min}} = 2\pi \times 19.7 \times 0.0225/50.3 = 55.4 \text{ mm} \tag{95}$$

A lever arm of 56 mm or longer makes the velocity constraint non-binding.

Config 2: prescribed base motion. For Config 2 the motor prescribes plate displacement $x_P(t) = A\sin(\omega_{n,1}t)$. The minimum plate amplitude for snap is $A_{\text{min}} = 2\zeta_h d_s = 2.25\,\text{mm}$. The corresponding peak shaft velocity:

$$v_{\text{peak}} = \frac{A_{\text{min}}}{r} \cdot \omega_{n,1} = \frac{2.25\,\text{mm}}{50\,\text{mm}} \times 123.8\,\text{rad/s} = 5.57 \text{ rad/s} \ll \omega_{\text{max}} \tag{96}$$

Config 2 is within the velocity limit and requires $\tau = 2\zeta_h\, k_{\text{eff}}\, d_s\, r \approx 0.34\,\text{N·m}$ for a 50 mm lever arm, about 21% of rated torque.

4.3 Control Modes

The run mode is selected by writing parameter `run_mode` (index `0x7005`) via Communication Type 18, then enabling the motor with Type 3.

Table 4: RS05 control modes (parameter `run_mode`, index `0x7005`).

Value	Mode	Description
0	Operation (MIT-style)	$\tau_{\text{ref}} = K_d(v_{\text{ref}} - v) + K_p(\theta_{\text{ref}} - \theta) + t_{ff}$; set $K_p = K_d = 0$ for pure torque feedforward
1	Position PP	S-curve trajectory to target angle; no mid-move speed changes
2	Velocity	Speed loop with current limit
3	Current	Direct Iq command; no position or speed loop
5	Position CSP	Cyclic synchronous position; velocity-limited

For sinusoidal bistable excitation, **mode 0 (operation) or mode 3 (current)** are the applicable choices.

4.4 Commanding a Sinusoidal Torque via CAN

4.4.1 No Onboard Sine Generation

The motor firmware contains no "sine wave start" CAN command and no writable registers for `sine_amplitude` or `sine_frequency`. The "Sine Wave Test" panel in the MotorStudio PC debugger is entirely host-side: the PC application computes the sinusoidal setpoint at each tick and streams individual CAN frames to the motor. An embedded controller (e.g. ESP32) must do the same.

4.4.2 Option A — Operation Control Mode (Type 1 Frame)

The Type 1 frame packs five 16-bit values into a 29-bit extended CAN ID plus 8-byte data field:

Field	Bistable excitation setting
t_{ff} [0–65535] $\to$ $[-5.5, +5.5]$ N·m	$\tau_0 \sin(2\pi f_{n,1} t)$, scaled
Angle	0
Velocity	0
K_p	0
K_d	0

The motor responds with a Type 2 feedback frame (position, velocity, torque, temperature) after every Type 1 command.

4.4.3 Option B — Current Mode with `iq_ref` (Type 18)

1. Write `run_mode` = 3 via Type 18 (index `0x7005`).
2. Enable motor: send Type 3.
3. Each tick: write `iq_ref` (index `0x7006`, float, ± 11 A) via Type 18:

$$i_q(t) = \frac{\tau_0}{K_t} \sin(2\pi f_{n,1}\, t) \tag{97}$$

For $\tau_0 = 0.34\,\text{N·m}$ and $K_t = 0.94\,\text{N·m/A}$: $i_{q,0} = 0.36\,\text{A}$ — well within the $\pm 11\,\text{A}$ limit.

The speed controller is inactive in current mode; the motor delivers the commanded torque regardless of shaft velocity, subject only to back-EMF and the current hardware limit.

4.5 ESP32 Feasibility Assessment

Table 5: CAN bus load for two RS05 motors commanded at 500 Hz per motor.

Parameter	**Value**
CAN bus rate	1 Mbps
Frame size (extended)	≈ 110 bits
Frame transmission time	$\approx 110\,\mu$s
Frames/sec (2 motors $\times$ 500 Hz)	1000
CAN bus utilisation	$1000 \times 110\,\mu\text{s} = 11\%$
ESP32 loop period	1 ms (hardware timer, Core 1)
Timing jitter	$\pm$10–50 μs
Phase error at 19.7 Hz	$\leq 50\,\mu\text{s}/50.8\,\text{ms} \approx 0.1\%$
Motor active reporting	min 10 ms (100 Hz) per motor

The ESP32 is sufficient for this task. Key implementation notes:

- Disable WiFi/Bluetooth; pin the CAN interrupt and timer callback to Core 1 at high FreeRTOS priority to avoid latency spikes.
- Use `esp_timer_create` (not `vTaskDelay`) for the 1 ms tick; jitter is then $< 50\,\mu$s.
- Accumulate phase as a running sum ($\varphi \mathrel{+}= 2\pi f \Delta t$ each tick) rather than computing $\sin(2\pi f t_{\text{wall}})$ to avoid jitter corrupting the waveform.
- Snap detection: read position (Bytes 0–1 of Type 2 feedback) each reporting cycle; a position jump of $\geq d_s$ through the lever arm signals snap and the torque command should be zeroed.
- Enable CAN timeout protection: set `CAN_TIMEOUT` > 0 so the motor enters reset mode if CAN communication is lost (prevents runaway).

5 Optimization

5.1 Snap-Through Threshold

From linear resonance theory, the steady-state hook amplitude at $\omega = \omega_{n,1}$ is:

$$A_{\text{ss}} = \frac{F_0}{2\zeta_h\, k_{\text{eff},1}} \tag{98}$$

Snap-through occurs when $A_{\text{ss}} \geq d_s$, giving:

$$\boxed{F_{0,\min} = 2\,\zeta_h\, k_{\text{eff},1}\, d_s} \tag{99}$$

This is the *minimum force applied directly at the hook*. For base excitation (motor on plate), the motor force required is larger by a factor that depends on the plate dynamics and drive frequency.

5.2 Mass Actuation Force

The central design question for the Rolly deployment mechanism is: *for a given mass budget added to the robot, what is the maximum force that mass can deliver to the bistable hook?*

5.2.1 Proof-Mass Oscillator (Section 3.12)

A proof mass M_{pm} oscillating at frequency ω and amplitude Δ delivers inertial force $F = M_{pm}\,\Delta\,\omega^2$. At the velocity limit (eq. (90)):

$$\Delta_{\max}(\omega) = \frac{\omega_{\max}\, r}{\omega} \tag{100}$$

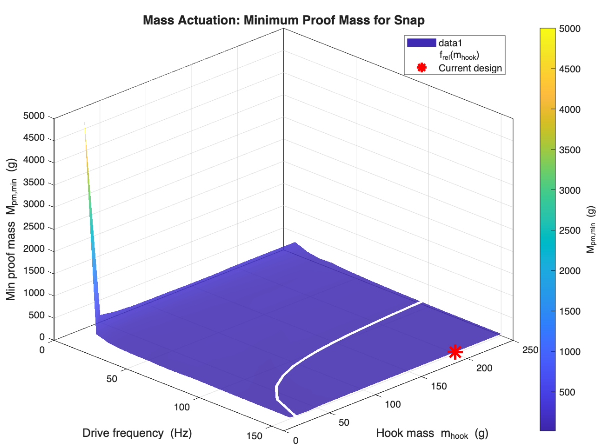
The maximum force per unit proof mass is:

$$\frac{F_{\max}}{M_{pm}} = \Delta_{\max}\,\omega^2 = \omega_{\max}\, r\,\omega \tag{101}$$

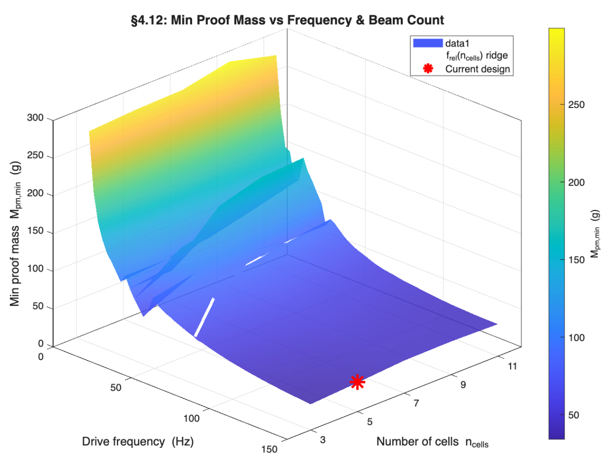
This grows *linearly with drive frequency*. Higher frequency delivers more force for the same proof mass — up to the limit imposed by the bistable dynamics (TMD anti-resonance at $\omega_{n,1}$; see Section 3.8). The coupled relative-mode resonance at ω_{rel} is the correct drive frequency:

$$\omega_{\text{rel}} = \sqrt{\frac{k_{\text{eff},1}}{M_{\text{red}}}}, \qquad M_{\text{red}} = \frac{m_h\, M_P}{m_h + M_P} \tag{102}$$

The minimum proof mass to snap at ω_{rel} is found by binary search over the nonlinear ODE.



(a) $M_{pm,\min}$ vs. drive frequency and hook mass m_h.



(b) $M_{pm,\min}$ vs. drive frequency and number of cells N.

Figure 6: 3-D design surfaces for the minimum required proof mass $M_{pm,\min}$ to achieve snap-through, swept over the drive frequency and two key design variables. Each surface is obtained by binary search over the nonlinear ODE at every grid point.

5.2.2 Config 2: Prescribed Base Motion (no added mass)

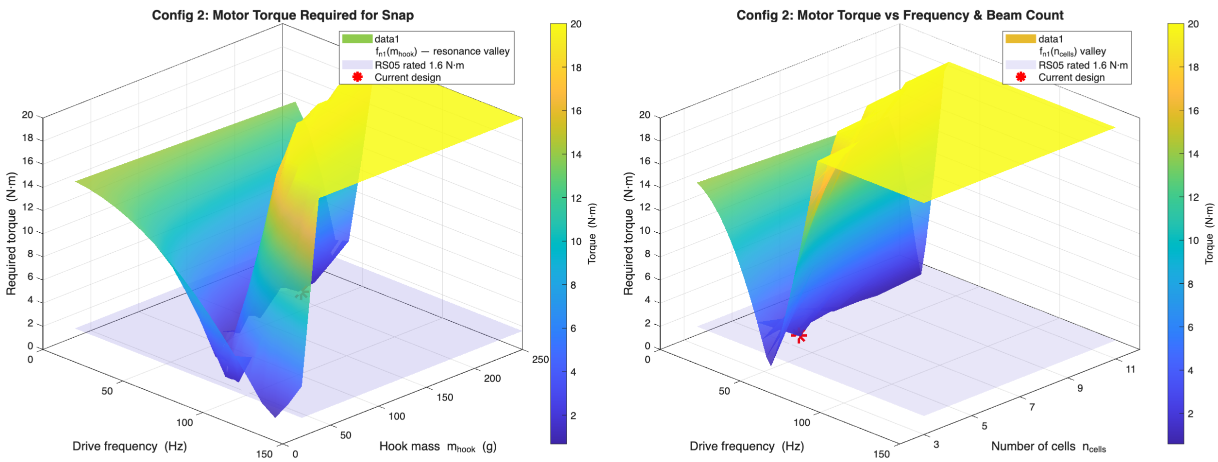
The motor prescribes the plate displacement $x_P(t) = A\sin(\omega_{n,1}t)$. The minimum plate amplitude for resonant snap is:

$$A_{\min} = 2\,\zeta_h\,d_s \tag{103}$$

The required motor torque at resonance ($\omega = \omega_{n,1}$):

$$\tau = m_h\,A_{\min}\,\omega_{n,1}^2\,r = 2\,\zeta_h\,k_{\text{eff},1}\,d_s\,r \tag{104}$$

This is *independent of hook mass m_h*. No additional hardware mass is required — the motor already present for locomotion is used.



(a) Required torque vs. drive frequency and hook mass m_h.

(b) Required torque vs. drive frequency and number of cells N.

Figure 7: 3-D design surfaces for the required motor torque under Configuration 2 (prescribed base motion) for snap-through, swept over the drive frequency and two key design variables.

5.3 Stacked-Cell Scaling (N cells)

Adding bistable cells in series scales the mechanism as follows:

$$d_f = N\,d_{f,1} \quad \text{(stroke grows linearly)} \tag{105}$$

$$k_{\text{eff},1} \propto \frac{1}{N} \quad \text{(stiffness drops)} \tag{106}$$

$$f_{n,1} \propto \frac{1}{\sqrt{N}} \quad \text{(natural frequency drops)} \tag{107}$$

$$F_{0,\min} \approx \text{const} \quad \text{(independent of } N\text{, see eq. (99))} \tag{108}$$

The last result follows because $k_{\text{eff},1} \propto 1/N$ and $d_s \propto N$, so their product $k_{\text{eff},1}\,d_s$ is constant.

The post-snap potential energy available to accelerate the hook to State 2:

$$\Delta E_2 = V(d_s) - V(d_f) \tag{109}$$

grows with N (deeper well, larger stroke). This is the *impact energy* available when the hook collides with a target object at State 2.

5.4 Impact Force at State 2

When the hook arrives at $x_{\text{rel}} = d_f$ it has kinetic energy accumulated from two sources:

1. **Kinetic energy at saddle crossing** KE_{saddle} (from resonant build-up).
2. **Potential energy released** by the bistable spring from saddle to State 2: ΔE_2.

The hook velocity at impact (ignoring damping over the short snap transit):

$$v_{\text{impact}} = \sqrt{v_{\text{saddle}}^2 + \frac{2\,\Delta E_2}{m_h}} \tag{110}$$

At the minimum snap threshold $v_{\text{saddle}} \approx 0$, so $v_{\text{impact}} \approx \sqrt{2\,\Delta E_2/m_h}$.

The peak force on the target object (Hertz contact model):

$$F_{\text{peak}} = v_{\text{impact}}\,\sqrt{k_{\text{contact}}\,m_h} \tag{111}$$

where k_{contact} is the contact stiffness. The impact force scales as $\sqrt{k_{\text{contact}}}$: steel-on-steel ($k \sim 10^{10}\,\text{N/m}$) gives $\sim 3000\times$ more peak force than rubber-on-rubber ($k \sim 10^4\,\text{N/m}$) for the same impact velocity.

The impact momentum (relevant for deforming targets):

$$p = m_h\,v_{\text{impact}} = \sqrt{2\,m_h\,\Delta E_2} \tag{112}$$

grows as $\sqrt{m_h}$ — heavier hooks deliver more momentum.

5.5 Design Sensitivity

Table 6: Effect of increasing each design variable on key performance metrics. $\uparrow$ = increases, $\downarrow$ = decreases, $\leftrightarrow$ = unchanged.

Variable	$f_{n,1}$	d_f	ΔE_2	$F_{0,\min}$	$M_{pm,\min}$	τ_{C2}
$\uparrow m_h$	$\downarrow$	$\leftrightarrow$	$\leftrightarrow$	$\leftrightarrow$	see note	$\leftrightarrow$
$\uparrow N$	$\downarrow$	$\uparrow$	$\uparrow$	$\leftrightarrow$	$\downarrow$	$\leftrightarrow$
$\uparrow \zeta_h$	$\leftrightarrow$	$\leftrightarrow$	$\leftrightarrow$	$\uparrow$	$\uparrow$	$\uparrow$
$\uparrow d_{s,\text{frac}}$	$\downarrow$	$\leftrightarrow$	$\downarrow$	$\uparrow$	$\uparrow$	$\uparrow$

Note on m_h: $M_{pm,\min}$ depends on the full coupled dynamics. Heavier hooks lower $f_{n,1}$ and f_{rel}, shifting the optimal drive frequency and changing $M_{pm,\min}$ non-monotonically. The 3-D surface $M_{pm,\min}(f,\,m_h)$ from the MATLAB simulation reveals the full dependence.

To **maximise impact energy**: increase N (more cells, deeper well).

To **minimise required actuator mass** (M_{pm}): drive near f_{rel} (avoid TMD anti-resonance), increase N (lower $f_{n,1}$ and f_{rel}, higher F/M from eq. (101)).

To **minimise motor torque** (Config 2): drive at $f_{n,1}$; torque is independent of m_h and N (both cancel in eq. (104)).

To **ensure bistability**: maintain $Q = h/t \gtrsim 2.35$ and use a double-beam layout to suppress mode 2.

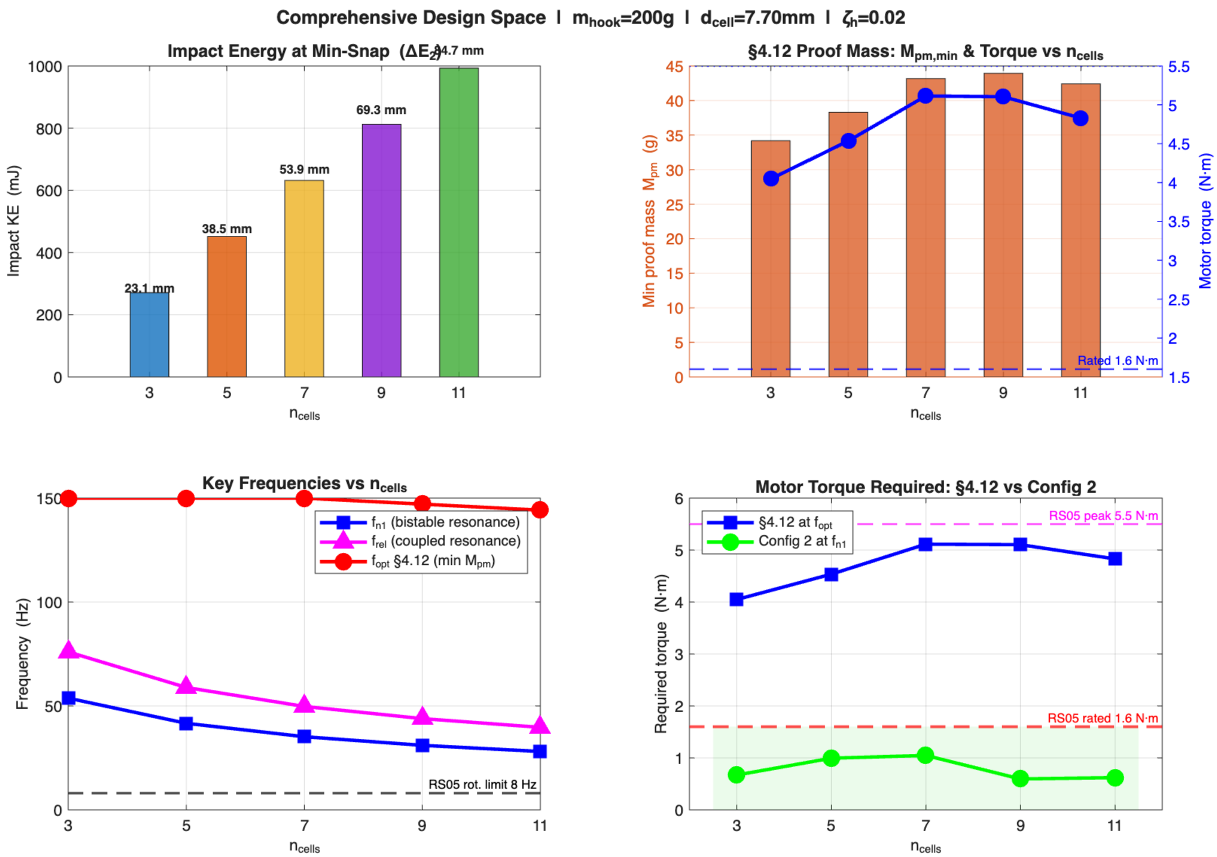


Figure 8: Comprehensive design space: key performance metrics (natural frequency $f_{n,1}$, energy barrier ΔE_1, impact energy ΔE_2, minimum required force $F_{0,\min}$, and minimum proof mass $M_{pm,\min}$) as a function of the number of stacked cells N. All curves are normalised to their single-cell values.

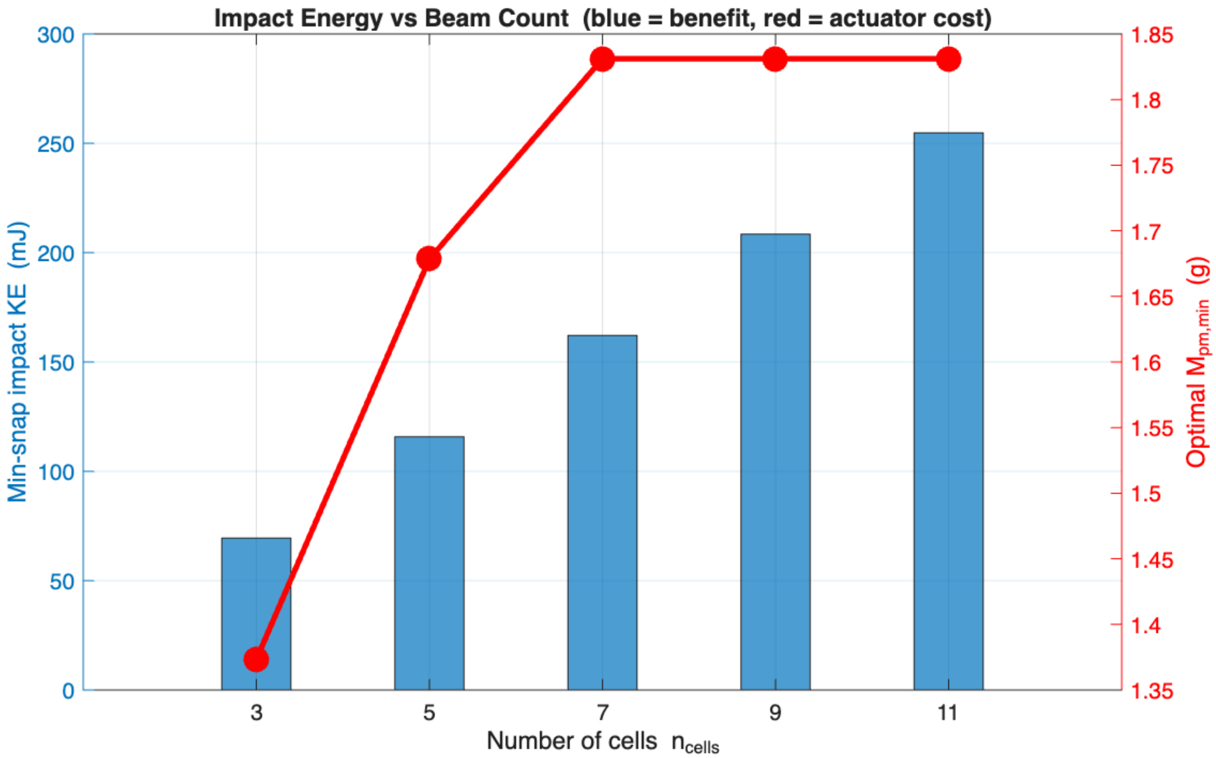


Figure 9: Impact energy ΔE_2 and actuation cost metrics as a function of the number of stacked cells N. Impact energy grows with N (deeper potential well, longer stroke), while the minimum proof mass $M_{pm,\min}$ decreases, making additional cells doubly beneficial.

6 Nomenclature

Table 7: Symbol definitions used in this document and the MATLAB simulations.

Symbol	**Unit**	**Description**	**Typical value**
Simulation 2 — Curved-beam bistable			
E	Pa	Elastic modulus (ABS)	2300×10^{6}
t	m	Beam thickness	0.6×10^{-3}
h	m	Beam apex height	2.5×10^{-3}
l	m	Beam span	42×10^{-3}
Q	—	Apex-height-to-thickness ratio	h/t (target $\gtrsim 2.35$)
N	—	Number of stacked cells	5
d_f	m	Total stroke (N cells)	$N \times 8.25 \times 10^{-3}$
d_s	m	Saddle position	$0.545\, d_f$
A	N/m^3	Cubic polynomial coefficient	from eq. (8)
F_{push}	N	Peak push force (per cell)	12.55 (sim), 11.10 (exp)

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Symbol	Unit	Description	Typical value
F_{pop}	N	Valley force	from cubic fit
$k_{\text{eff},1}$	N/m	Stiffness at State 1	$A\, d_s\, d_f$
$k_{\text{eff},2}$	N/m	Stiffness at State 2	$A\, d_f(d_f - d_s)$
$f_{n,1}$	Hz	Natural freq. at State 1	$\omega_{n,1}/(2\pi)$
$f_{n,2}$	Hz	Natural freq. at State 2	$\omega_{n,2}/(2\pi)$
m_1	kg	Main mass (TMD model)	0.150
m_2	kg	TMD mass	0.01875
k_{main}	N/m	Main spring stiffness	7253 (35 Hz)
E_r	Pa	Ruler elastic modulus	210×10^9
Section 3 — Base excitation model			
x_P	m	Absolute plate displacement	(simulated)
x_{rel}	m	Hook displacement relative to plate	(simulated)
M_P	kg	Plate + bistable assembly mass (motor excluded)	0.200
f_P	Hz	Plate support natural frequency	set to $f_{n,1}$
ζ_P	—	Plate structural damping ratio	0.02
k_s	N/m	Plate support stiffness	$M_P(2\pi f_P)^2$
c_s	N·s/m	Plate support damping	$2\zeta_P M_P(2\pi f_P)$
F_0	N	Motor force amplitude at plate	$\tau_{\text{out}}/r_{\text{arm}}$
τ_{out}	N·m	Measured RS05 output torque (from T-N curve)	≤ 5.5
r_{arm}	m	Coupling moment arm	0.010
n_g	—	RS05 planetary gear ratio	7.75
J_{eff}	kg·m^2	Reflected rotor inertia at output	$J_{\text{rotor}}\, n_g^2$
f_{cut}	Hz	Gearbox low-pass cut-off frequency	$\frac{1}{2\pi}\sqrt{k_{\text{gear}}/J_{\text{eff}}}$
f_{mesh}	Hz	Gear-mesh frequency	$n_g f_{\text{rotor}} N_{\text{teeth}}$
n	—	Detuning ratio $f_P/f_{n,1}$	1 (tuned)
Q_P	—	Plate quality factor $1/(2\zeta_P)$	25
Q_h	—	Hook quality factor $1/(2\zeta_h)$	10
Section 5 — Optimization and mass actuation			
N	—	Number of bistable cells in series	5
f_{rel}	Hz	Coupled relative-mode resonance $= \sqrt{k_{\text{eff},1}/M_{\text{red}}}/(2\pi)$	22
M_{red}	kg	Reduced mass $= m_h M_P/(m_h + M_P)$	0.133
M_{pm}	kg	Proof mass (oscillating on motor shaft)	optimised
Δ	m	Proof-mass oscillation amplitude	velocity-limited
$\omega_{\max}$	rad/s	RS05 max shaft angular velocity (480 RPM)	50.3
$M_{pm,\min}$	kg	Minimum proof mass to snap at given (f, N)	from sweep
$A_{\min}$	m	Minimum plate amplitude for Config 2 snap	$2\zeta_h d_s$

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Symbol	Unit	Description	Typical value
τ_{C2}	N·m	Config 2 motor torque $= 2\zeta_h k_{\text{eff}} d_s r$	0.34
ΔE_2	J	Post-snap potential energy released (saddle to State 2)	grows with N
v_{impact}	m/s	Hook velocity at State 2 impact	$\sqrt{2\Delta E_2/m_h}$
k_{contact}	N/m	Target contact stiffness (Hertz)	10^4–10^{10}
F_{peak}	N	Peak impact force $= v_{\text{impact}}\sqrt{k_c m_h}$	—

References

References

[1] A. H. Nayfeh and D. T. Mook, *Nonlinear Oscillations*, Wiley-Interscience, New York, 1979.

[2] J. P. Den Hartog, *Mechanical Vibrations*, 4th ed., McGraw-Hill, New York, 1956.

[3] J. Qiu, A. H. Slocum, J. H. Lang, R. Struempler, M. P. Brenner, and J. Li, *Bistable Actuation Techniques, Mechanisms, and Applications*, U.S. Patent Application Publication US 2003/0029705 A1, Feb. 2003.

[4] J. Qiu, J. H. Lang, and A. H. Slocum, "A Curved-Beam Bistable Mechanism," *Journal of Microelectromechanical Systems*, vol. 13, no. 2, pp. 137–146, 2004. https://doi.org/10.1109/JMEMS.2004.825308

[5] A. Zolfagharian, F. Demoly, M. Lakhi, B. Rolfe, and M. Bodaghi, "Bistable Mechanisms 3D Printing for Mechanically Programmable Vibration Control," *Advanced Engineering Materials*, vol. 28, no. 9, p. 2402233, 2026. https://doi.org/10.1002/adem.202402233

[6] Y. Liu, J. Zhang, D. Pan, Z. Wu, and Q. Wang, "Resonant Actuation Based on Dynamic Characteristics of Bistable Laminates," *Machines*, vol. 11, no. 3, p. 318, 2023. https://doi.org/10.3390/machines11030318

[7] RobStride, *RS05 User Manual*, Rev. 260428, 2026.